STACKS, MODULI OF CURVES: LEMMA-CONTRACT-BASIC

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2986-2995 of sources/stacks-moduli-curves/pinned-original-source.tex; proof span is lines 2997-3124, the complete author proof (126 body lines).

Editorial note. Navigation stubs: 14 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 0 of queue_moduli-curves.json (pb=5310, stmt 2986-2995, proof 2997-3124).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 02KH: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 3124–3124. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0E3X: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 3124–3124. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DSS: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 655–658. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01ZM: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 3124–3124. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 081F: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 3124–3124. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07QR: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 3124–3124. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:moduli-curves.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 07PX: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 3124–3124. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 05KS: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 3124–3124. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0D58: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 3124–3124. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07QY: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 3124–3124. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07SK: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 3124–3124. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0A4Z: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 3124–3124. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0ADD: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 3124–3124. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0CMV: sources/stacks-moduli-curves/pinned-original-source.tex, source lines 3124–3124. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to S$ be a family of curves. Let $s \in S$ be a point. Let $h_0 : X_s \to Y_0$ be a morphism to a proper scheme Y_0 over $\kappa(s)$ such that $h_{0,*}\mathcal{O}_{X_s} = \mathcal{O}_{Y_0}$ and $R^1 h_{0,*}\mathcal{O}_{X_s} = 0$. Then there exist an elementary étale neighbourhood $(U, u) \to (S, s)$, a family of curves $Y \to U$, and a morphism $h : X_U \to Y$ over U whose fibre in u is isomorphic to h_0.*

Proof. We first do some reductions; we urge the reader to skip ahead. The question is local on S, hence we may assume S is affine. Write $S = \lim S_i$ as a cofiltered limit of affine schemes S_i of finite type over $\mathbf{Z}$. For some i we can find a family of curves $X_i \to S_i$ whose base change is $X \to S$. This follows from Lemma 0DSS and Limits of Stacks, Lemma 0CMV. Let $s_i \in S_i$ be the image of s. Observe that $\kappa(s) = \operatorname{colim} \kappa(s_i)$ and that X_s is a scheme (Spaces over Fields, Lemma 0ADD). After increasing i we may assume there exists a morphism $h_{i,0} : X_{i,s_i} \to Y_i$ of finite type schemes over $\kappa(s_i)$ whose base change to $\kappa(s)$ is h_0, see Limits, Lemma 01ZM. After increasing i we may assume Y_i is proper over $\kappa(s_i)$, see Limits, Lemma 081F. Let $g_{i,0} : Y_0 \to Y_{i,0}$ be the projection. Observe that this is a faithfully flat

morphism as the base change of $\mathrm{Spec}(\kappa(s)) \to \mathrm{Spec}(\kappa(s_i))$. By flat base change we have

$$h_{0,*}\mathcal{O}_{X_s} = g_{i,0}^* h_{i,0,*}\mathcal{O}_{X_{i,s_i}} \quad \text{and} \quad R^1 h_{0,*}\mathcal{O}_{X_s} = g_{i,0}^* R h_{i,0,*}\mathcal{O}_{X_{i,s_i}}$$

see Cohomology of Schemes, Lemma 02KH. By faithful flatness we see that $X_i \to S_i$, $s_i \in S_i$, and $X_{i,s_i} \to Y_i$ satisfies all the assumptions of the lemma. This reduces us to the case discussed in the next paragraph.

Assume S is affine of finite type over $\mathbf{Z}$. Let $\mathcal{O}_{S,s}^h$ be the henselization of the local ring of S at s. Observe that $\mathcal{O}_{S,s}^h$ is a G-ring by More on Algebra, Lemma 07QR and Proposition 07PX. Suppose we can construct a family of curves $Y' \to \mathrm{Spec}(\mathcal{O}_{S,s}^h)$ and a morphism

$$h' : X \times_S \mathrm{Spec}(\mathcal{O}_{S,s}^h) \longrightarrow Y'$$

over $\mathrm{Spec}(\mathcal{O}_{S,s}^h)$ whose base change to the closed point is h_0. This will be enough. Namely, first we use that

$$\mathcal{O}_{S,s}^h = \mathrm{colim}_{(U,u)} \mathcal{O}_U(U)$$

where the colimit is over the filtered category of elementary étale neighbourhoods (More on Morphisms, Lemma 05KS). Next, we use again that given Y' we can descend it to $Y \to U$ for some U (see references given above). Then we use Limits, Lemma 01ZM to descend h' to some h. This reduces us to the case discussed in the next paragraph.

Assume $S = \mathrm{Spec}(\Lambda)$ where $(\Lambda, \mathfrak{m}, \kappa)$ is a henselian Noetherian local G-ring and s is the closed point of S. Recall that the map

$$\mathcal{D}ef_{X_s \to Y_0} \to \mathcal{D}ef_{X_s}$$

is an equivalence, see Deformation Problems, Lemma 0E3X. (This is the only important step in the proof; everything else is technique.) Denote $\Lambda^\wedge$ the $\mathfrak{m}$-adic completion. The pullbacks X_n of X to $\Lambda/\mathfrak{m}^{n+1}$ define a formal object ξ of $\mathcal{D}ef_{X_s}$ over $\Lambda^\wedge$. From the equivalence we obtain a formal object ξ' of $\mathcal{D}ef_{X_s \to Y_0}$ over $\Lambda^\wedge$. Thus we obtain a huge commutative diagram

$$\begin{array}{ccccccc}
\ldots & \longrightarrow & X_n & \longrightarrow & X_{n-1} & \longrightarrow \ldots \longrightarrow & X_s \\
 & & \downarrow & & \downarrow & & \downarrow \\
\ldots & \longrightarrow & Y_n & \longrightarrow & Y_{n-1} & \longrightarrow \ldots \longrightarrow & Y_0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
\ldots & \longrightarrow & \mathrm{Spec}(\Lambda/\mathfrak{m}^{n+1}) & \longrightarrow & \mathrm{Spec}(\Lambda/\mathfrak{m}^n) & \longrightarrow \ldots \longrightarrow & \mathrm{Spec}(\kappa)
\end{array}$$

The formal object (Y_n) comes from a family of curves $Y' \to \mathrm{Spec}(\Lambda^\wedge)$ by Quot, Lemma 0D58. By More on Morphisms of Spaces, Lemma 0A4Z we get a morphism $h' : X_{\Lambda^\wedge} \to Y'$ inducing the given morphisms $X_n \to Y_n$ for all n and in particular the given morphism $X_s \to Y_0$.

To finish we do a standard algebraization/approximation argument. First, we observe that we can find a finitely generated Λ-subalgebra $\Lambda \subset A \subset \Lambda^\wedge$, a family of curves $Y'' \to \mathrm{Spec}(A)$ and a morphism $h'' : X_A \to Y''$ over A whose base change to $\Lambda^\wedge$ is h'. This is true because $\Lambda^\wedge$ is the filtered colimit of these rings A and we can argue as before using that *Curves* is locally of finite presentation (which gives us Y'' over A by Limits of Stacks, Lemma 0CMV) and using Limits of Spaces, Lemma 07SK to descend h' to some h''. Then we can apply the approximation property for G-rings (in the form of Smoothing Ring Maps, Theorem 07QY) to find a map $A \to \Lambda$ which induces the same map $A \to \kappa$ as we obtain from $A \to \Lambda^\wedge$. Base changing h'' to Λ the proof is complete. $\square$